

Tae Hoon Yang

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EDUCATION

Stanford University, Stanford, CA

M.S. in Computer Science (Specialization in AI), GPA: 4.1/4.0

Expected Jun 2026

Relevant Courses: Deep Learning for Computer Vision, Deep RL, Sensorimotor Learning for Embodied Agents, Machine Learning, AI Principles, Robot Perception, Decision Making under Uncertainty

Northeastern University, Boston, MA

B.S. in Computer Engineering and Minor in Robotics, Summa Cum Laude

May 2024

Relevant Courses: Computer Vision, Robotics Sensing & Navigation, Mobile Robotics, Robot Dynamics & Control

Activities: Korean Computer Science Community (Founder), Med Finder Startup (Engineering Lead), Husky Startup Challenge (3rd Place Winner), Eta Kappa Nu, Tau Beta Pi, IEEE

RESEARCH & WORK EXPERIENCE

ToddlerBot Team – Stanford University, Stanford, CA

Dec 2024 – Present

- Develop a unified locomotion policy under Stanford's TML and REAL labs to enable ToddlerBot, a low-cost open-source humanoid robot, to walk, crawl, and climb using egocentric vision.
- Build skill-specific expert policies and apply policy distillation to create a single robust controller, improving motion stability and enabling seamless behavior transitions across diverse terrains.

Autonomy & Intelligence Lab – Northeastern University, Boston, MA

Jan 2024 – Aug 2024

- Configured Unitree Go1 robot dog with 3D LiDAR sensor fixture and ROS2 environment on robot's Jetson board.
- Integrated state estimation from DLIOM (Direct LiDAR Inertial Odometry and Mapping) with ROS2 Nav2.
- Achieved navigation with custom nodes that convert velocities from Nav2 to robot's high-level SDK commands.

Hardware Automation Engineering Intern – iRobot, Bedford, MA

Jul 2023 – Dec 2023

- Implemented image processing (Blurring, Filtering, Edge Detection, Binary Threshold, and Structural Similarity Index Measure) to grade dirtiness of tiles images in Python and C#.
- Evaluated robot's motion characteristics (parallelism, straightness, coverage, etc.) analyzing Qualisys motion capture data.
- Created power supply C# libraries by converting SCPI commands from manuals and establishing USB or RS-232 connections.

TEACHING EXPERIENCE

EECE 2160 Embedded Design: Enabling Robotics TA – Northeastern University, Boston, MA

Fall 2022, Spring 2024

- Instructed students during labs covering FPGA, Quartus Prime, Linux, C++, and robotic arm.
- Graded assignments for Digital Logic Design and C++, focusing on pointers, dynamic memory, data structures, and OOP.

EECE 2140 Computing Fundamentals TA – Northeastern University, Boston, MA

Spring 2023

- Conducted office hours to assist students with Python programming covering syntaxes, DSA, OOP, NumPy, unit testing, etc.
- Graded Python assignments by designing and implementing comprehensive rubrics.

ENGINEERING PROJECTS

Enhancing Monocular Metric Depth Estimation through Learned Adaptive Scaling

Jan 2025 – Mar 2025

- Identified and quantified global metric scaling errors from pretrained monocular depth models due to dataset bias.
- Trained a lightweight CNN to predict scene-specific scaling factors, improving depth accuracy without retraining the base model.

RL-Based Safe Autonomous Driving at Uncontrolled Intersections

Oct 2024 – Dec 2024

- Designed and implemented a simulation environment for autonomous vehicle navigation at an uncontrolled intersection using SUMO, incorporating MDP and POMDP frameworks.
- Implemented and trained Deep Q-Network (DQN) and Deep Recurrent Q-Network (DRQN) to enable optimal decision-making in high-dimensional and partially observable environments, accounting for vehicle interactions and hidden states.

UUV (Underwater Unmanned Vehicle) Localization - 2nd Place, Capstone Design Competition

Jan 2024 – Apr 2024

- Developed underwater communication system using transducers, GPS, and STM32 microcontrollers for UART, ADC, and DAC interfaces on Raspberry Pi.
- Implemented Bayesian inference and Markov Chain Monte Carlo (MCMC) methods for geolocation with noisy Time-of-Flight (ToF) measurements incorporating model uncertainties for robust position estimation.

Image to Robot Drawing Trajectory Converter

Jul 2023 – Aug 2023

- Undistorted images captured by robot's fisheye camera using its intrinsic parameters and distortion coefficients.
- Detected outlines of objects in images with cascade classifier to locate ROI of objects and Canny Edge Detection.
- Converted outlines to robot drawing trajectories by searching closest outline pixels and indicating non-drawing area.

Autonomous Reconnaissance Robot

Nov 2022 – Dec 2022

- Configured Turtlebot3 robot with 2D LiDAR, Pi camera, SBC, and OpenCR board installing ROS and Turtlebot3 packages.
- Performed Frontier-Based Exploration using Gmapping SLAM in closed testbed to generate occupancy grid map.
- Transformed coordinates of AprilTags detected by Pi Camera from camera frame to map frame to display tag poses on map.

SKILLS

Programming: C/C++, C#, Python, MATLAB | **Development Tools:** ROS, ROS2, Linux, Git, Docker | **Hardware &**

Electronics: 2D/3D LiDAR, Fisheye/Stereo/RGB-D Camera, 9-DOF IMU, GNSS/RTK, Raspberry Pi, Nvidia Jetson, STM32